

Product Performance Analysis

Big Data Project Report

Project Team

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2024 – 2025

Section 1: Business Area

Business Area: Sales Performance and Promotion Effectiveness Analysis

This schema is a classic star schema centered on the **FACT_PRODUCT_SALES** table, supported by dimensions such as Date, Customer, Product, Territory, and Promotion. In addition, the **FACT_PROMOTION_COVERAGE** table extends the model by capturing which products are eligible for promotions. The primary business area being analyzed is:

1) Core Objective

Evaluate product sales performance and promotion coverage effectiveness across multiple dimensions to support operational and strategic decision-making, including identifying gaps between promoted products and actual sales.

2) Key Analytical Focus Areas

a. Revenue and Profitability

Metrics: GrossSalesAmount, NetSalesAmount, MarginAmount, StandardCostAmount

Goal: Identify high-margin products, cost inefficiencies, and revenue drivers.

b. Customer Behavior

Dimensions: Customer type, geography, account details.

Goal: Segment customers (e.g., store vs individual), analyze purchasing patterns, and track customer lifetime value.

c. Time-Based Trends

Dimensions: Date (day, month, quarter, fiscal year).

Goal: Detect seasonality, sales cycles, and trends over time.

d. Product Performance

Dimensions: Product category, subcategory, attributes (color, size, style).

Goal: Identify top-performing products and underperforming SKUs.

e. Geographic Performance

Dimensions: Territory (region, country).

Goal: Compare sales across regions and optimize market strategies.

f. Promotion Effectiveness and Coverage Gap Analysis

Fact Tables: FACT_PRODUCT_SALES, FACT_PROMOTION_COVERAGE

Dimensions: Promotion type, discount percentage, duration, product attributes.

Goals:

- Measure the impact of promotions on sales volume and revenue.
- Compare products included in promotions (coverage) vs products actually sold.

Identify:

- Promoted products that generated sales.
- Promoted products that did not sell (coverage gap).
- Products sold without promotions.

Analytical Value:

- Detect ineffective promotions and wasted discounting.
- Identify products that require better targeting or pricing strategies.

- Quantify the proportion of promoted inventory that fails to convert into sales.

Conceptual Model Extension

The inclusion of **FACT_PROMOTION_COVERAGE** enables a set-based analysis:

- **Promotion set** — eligible products
- **Sales set** — actual transactions
- **Difference** → unsold promoted products

This approach aligns with dimensional modeling practices for advanced analytics.

Section 2: Business Questions

The following 20 business questions are addressed through MDX queries executed against the AdventureWorksDW_View cube. Each question targets a specific analytical dimension of the sales and promotion model.

Q1 — Annual Sales Performance Overview

Business Question: How do gross sales, discounts applied, net sales, and profit margin compare across different order years?

Objective: Track year-over-year revenue growth, discount impact, and margin evolution to assess overall financial health.

Q2 — Quarterly Net Sales Trends

Business Question: How does net sales performance fluctuate across quarters, and which quarters consistently outperform others?

Objective: Detect seasonal patterns and cyclical peaks in revenue to guide quarterly planning and resource allocation.

Q3 — Monthly Net Sales Patterns

Business Question: What month-by-month net sales patterns exist, and are there identifiable peak or trough months?

Objective: Identify intra-year seasonality for inventory planning, promotional timing, and sales forecasting.

Q4 — Product Category Performance by Year

Business Question: Which product categories drive the most net sales, and how has each category's contribution changed year over year?

Objective: Evaluate category-level growth trajectories to prioritize investment and identify declining segments.

Q5 — Top 10 Products by Profit Margin

Business Question: Which 10 individual products generate the highest profit margin?

Objective: Identify high-value products to protect pricing, prioritize production, and inform premium positioning strategies.

Q6 — Top 10 Products by Net Sales Revenue

Business Question: Which 10 products generate the highest net sales revenue?

Objective: Pinpoint the top revenue contributors to focus marketing spend and ensure consistent stock availability.

Q7 — Sales and Margin by Product Subcategory

Business Question: How do order quantity, net sales, and margin compare across all product subcategories?

Objective: Identify high-volume but low-margin subcategories versus high-margin but low-volume ones to balance the product portfolio.

Q8 — Revenue and Margin by Customer Type

Business Question: How do net sales and profit margin differ between customer types (e.g., store vs individual)?

Objective: Segment customer profitability to tailor pricing models, loyalty programs, and B2B vs B2C channel strategies.

Q9 — Top 10 Customers by Net Sales

Business Question: Who are the top 10 customers ranked by net sales revenue?

Objective: Identify key accounts for customer retention efforts, VIP pricing, and relationship management prioritization.

Q10 — Sales and Margin by Country

Business Question: How do net sales and profit margin vary across different countries?

Objective: Assess country-level profitability to guide market entry, pricing localization, and regional investment decisions.

Q11 — Sales and Margin by Region Group

Business Question: How do net sales and profit margin compare across broader region groups?

Objective: Compare macro-level geographic performance to support strategic expansion planning and regional budget allocation.

Q12 — Promotion Performance by Promotion Type

Business Question: How do gross sales, discount amounts, net sales, and margin compare across different promotion types?

Objective: Evaluate which promotion types deliver the best revenue and margin outcomes versus those that erode profitability through excessive discounting.

Q13 — Promotion Performance by Promotion Category

Business Question: How do promotion categories differ in their impact on gross sales, discounts, net sales, and margin?

Objective: Identify which broader promotion categories (e.g., seasonal, clearance, loyalty) are most effective and which generate unjustified discount costs.

Q14 — Promotion Type Performance Over Time

Business Question: How has the net sales performance of each promotion type evolved across different order years?

Objective: Detect whether certain promotions are gaining or losing effectiveness over time, enabling strategic renewal or discontinuation decisions.

Q15 — Order Quantity and Discount by Promotion Type

Business Question: Which promotion types drive the highest order quantities, and how much discount cost is incurred per promotion type?

Objective: Assess the volume-to-cost efficiency of each promotion type to determine whether the volume uplift justifies the discount investment.

Q16 — Net Sales by Day of Week

Business Question: On which days of the week are net sales highest, and is there a discernible weekday vs weekend purchasing pattern?

Objective: Optimize staffing, promotions, and logistics scheduling by understanding intra-week demand distribution.

Q17 — Net Sales and Margin by Fiscal Year

Business Question: How do net sales and profit margin trend across fiscal years?

Objective: Align financial reporting with fiscal periods to provide budget holders and executives with performance data matched to planning cycles.

Q18 — Top 5 Territories by Profit Margin

Business Question: Which 5 sales territories generate the highest profit margin?

Objective: Identify the most profitable territories to prioritize sales force deployment, targeted campaigns, and territory-level investment.

Q19 — Top 5 Promotion Types by Net Sales

Business Question: Which 5 promotion types contribute the most to net sales revenue?

Objective: Focus promotion budgets on the most revenue-generating promotion types and reduce spending on underperforming ones.

Q20 — Year-over-Year Sales Comparison Matrix

Business Question: How do gross sales, net sales, and profit margin compare side by side across all available order years?

Objective: Provide a holistic multi-year view for executive dashboards, enabling rapid identification of growth trends, revenue gaps, and margin compression across time.

Section 3: Source ERD / Schema (Interested Tables)

Source Database Overview

The source operational database is **AdventureWorks OLTP**, a relational database management system with around 70 normalized tables organized into schemas like **Production** (products/inventory), **Sales** (orders/customers), **Purchasing** (vendors), **Person** (contacts), and **HumanResources** (employees).

Selected Source Tables

The following 14 tables were selected from the AdventureWorks OLTP source as the foundation for the dimensional model. Each table contributes specific attributes to either the fact tables or dimension tables in the data warehouse.

#	Table Name	Primary Key	Description	DW Role
1	Sales.SalesOrderDetail	SalesOrderDetailID	Stores individual line items of each sales order, including quantity ordered, unit price, and discount applied. This is the core transactional table that populates the fact table.	Core Fact Source
2	Sales.SalesOrderHeader	SalesOrderID	Stores the master record for each sales order, including order date, ship date, due date, customer reference, and territory. Links to SalesOrderDetail via SalesOrderID.	Fact / Date / Territory Bridge
3	Production.Product	ProductID	Contains product details such as name, product number, color, size, style, standard cost, list price, and sell start/end dates. Feeds the product dimension.	Product Dimension
4	Production.ProductSubcategory	ProductSubcategoryID	Classifies products into subcategories (e.g., Road Bikes, Helmets). Links upward to ProductCategory and downward to Product.	Product Dimension
5	Production.ProductCategory	ProductCategoryID	Represents the top-level product classification (e.g., Bikes, Accessories, Clothing). Forms the highest level of the product hierarchy.	Product Dimension
6	Sales.Customer	CustomerID	Identifies each customer and links them to either a Person or a Store, along with their associated territory. Feeds the customer dimension.	Customer Dimension

#	Table Name	Primary Key	Description	DW Role
7	Person.Person	BusinessEntityID	Stores individual person details including first name, last name, and person type. Used to resolve individual customer identities from Sales.Customer.	Customer Dimension
8	Sales.Store	BusinessEntityID	Represents business/store customers with a store name and sales person reference. Used alongside Person.Person to fully describe the customer dimension.	Customer Dimension
9	Person.Address	AddressID	Contains physical address details including city and postal code. Used to enrich the geography dimension alongside StateProvince and CountryRegion.	Geography Dimension
10	Person.StateProvince	StateProvinceID	Stores state and province information linked to a country and territory. Forms the intermediate level of the geography hierarchy.	Geography Dimension
11	Person.CountryRegion	CountryRegionCode	Stores country and region names. Forms the top level of the geography hierarchy.	Geography Dimension
12	Sales.SalesTerritory	TerritoryID	Defines named sales territories grouped by country and region group (e.g., Northwest, Europe). Feeds the territory dimension for geographic performance analysis.	Territory Dimension

#	Table Name	Primary Key	Description	DW Role
13	Sales.SpecialOffer	SpecialOfferID	Describes each promotion or special offer including its type, category, discount percentage, and active date range. Feeds the promotion dimension.	<i>Promotion Dimension</i>
14	Sales.SpecialOfferProduct	SpecialOfferID, ProductID (composite)	A junction table linking promotions to eligible products. This table directly feeds the factless FACT_PROMOTION_COVERAGE table to capture promotion eligibility regardless of actual sales.	<i>Promotion Coverage Fact</i>

Note: Tables 1–2 are the primary transactional sources for FACT_PRODUCT_SALES. Table 14 (Sales.SpecialOfferProduct) is the exclusive source for FACT_PROMOTION_COVERAGE, enabling set-based coverage gap analysis independent of actual sales activity.

Source ERD Schema

https://drive.google.com/file/d/1Md8st8Vps62-gO5YOiJHKCSNipZc7lx3/view?usp=drive_link

Section 4: Dimensional Model Design (Star Schema)

1. Business Purpose

The model shall enable analysis of:

- Sales performance over time
- Product-level revenue, cost, and margin
- Customer purchasing behavior
- Sales by geography or territory
- Promotion usage and promotion coverage by product

2. Fact Tables

2.1 FACT_PRODUCT_SALES

This fact table shall store transactional sales data at the lowest business event level.

Grain

One row per sales order detail line. In practical terms: **one product line item in a sales order.**

- Sales order
- Product
- Customer
- Territory
- Promotion
- Order date / ship date / due date

Foreign Keys

Foreign Key Column	References Dimension
OrderDateKey	DIM_DATE
ShipDateKey	DIM_DATE
DueDateKey	DIM_DATE
ProductKey	DIM_PRODUCT
CustomerKey	DIM_CUSTOMER
TerritoryKey	DIM_TERRITORY
PromotionKey	DIM_PROMOTION

Degenerate Dimensions

Transactional identifiers retained for drill-through and traceability:

- SalesOrderID — order-level identifier

- SalesOrderDetailID — line-item identifier

Measures

Measure	Definition
OrderQuantity	Number of units ordered
UnitPrice	Per-unit price (semi-descriptive; not fully additive)
GrossSalesAmount	Quantity × unit price before discount
DiscountAmount	Reduction applied due to promotion or discount
NetSalesAmount	Gross sales – discount
StandardCostAmount	Standard unit cost × quantity
MarginAmount	Net sales – standard cost

2.2 FACT_PROMOTION_COVERAGE

This factless fact table shall record which products are eligible for which promotions.

Grain: One row per product–promotion eligibility combination. No numeric measures are stored; counts are derived by aggregation.

Foreign Key Column	References Dimension
ProductKey	DIM_PRODUCT
PromotionKey	DIM_PROMOTION

Derived counts (via aggregation):

- Count of products covered by a promotion
- Count of promotions applicable to a product

3. Dimension Tables

3.1 DIM_DATE

Supports calendar and fiscal time analysis. Role-playing dimension reused as order date, ship date, and due date in FACT_PRODUCT_SALES.

Attribute	Description
DateKey	Surrogate primary key
FullDate	Complete calendar date
DayNum	Day number of the month
MonthNum	Month number (1–12)
MonthName	Full month name
QuarterNum	Quarter number (1–4)
QuarterName	Quarter label (e.g., Q1)
YearNum	Calendar year
YearQuarterKey	Numeric year-quarter key
YearQuarterName	Year-quarter label (e.g., 2013 Q1)
YearMonthKey	Numeric year-month key
YearMonthName	Year-month label (e.g., 2013 Jan)
WeekdayName	Day of the week name
IsWeekend	Boolean flag for weekend days
WeekNumber	ISO week number
FiscalYear	Fiscal year value
FiscalQuarter	Fiscal quarter value

3.2 DIM_CUSTOMER

Describes the customer placing the order. Implements **SCD Type 2** using effective start/end dates and a current-record flag — enabling reporting by the customer profile as it existed at the time of sale.

Attribute	Description
CustomerKey	Surrogate primary key
CustomerID	Business key from source
CustomerType	Individual or store customer
CustomerName	Derived display name
PersonFullName	Full name for individual customers
StoreName	Store name for business customers
AccountNumber	Account identifier
EmailPromotion	Email promotion preference flag
City	Customer city
StateProvince	State or province
CountryRegion	Country or region
EffectiveStartDate	SCD Type 2 start date
EffectiveEndDate	SCD Type 2 end date
IsCurrent	Flag for the active record

3.3 DIM_PRODUCT

Describes the product sold. Implements **SCD Type 2** via effective dating and a current indicator, enabling sales analysis by product attributes as they existed at the time of the transaction.

Attribute	Description
ProductKey	Surrogate primary key
ProductID	Business key from source
ProductName	Product display name
ProductNumber	Product SKU code
Color	Product color
Size	Product size
Class	Product class (e.g., H, M, L)
Style	Product style
ProductLine	Product line grouping
DaysToManufacture	Manufacturing lead time in days
StandardCost	Standard unit cost
ListPrice	Published list price
SubcategoryName	Product subcategory
CategoryName	Product category (top level)
SellStartDate	Date product became available for sale
SellEndDate	Date product was retired from sale
DiscontinuedDate	Date product was discontinued
EffectiveStartDate	SCD Type 2 start date
EffectiveEndDate	SCD Type 2 end date
IsCurrent	Flag for the active record

3.4 DIM_TERRITORY

Describes the sales territory or geographic region, supporting geographic and regional sales analysis.

Attribute	Description
TerritoryKey	Surrogate primary key
TerritoryID	Business key from source
TerritoryName	Named sales territory
CountryRegionCode	ISO country or region code
RegionGroup	Broader regional grouping (e.g., Europe, North America)

3.5 DIM_PROMOTION

Describes the promotion or special offer applied to the sale. Supports analysis of promotion effectiveness, discount behavior, and eligibility rules.

Attribute	Description
PromotionKey	Surrogate primary key
SpecialOfferID	Business key from source
PromotionDescription	Full description of the promotion
DiscountPct	Discount percentage applied
PromotionType	Type of promotion (e.g., Volume Discount)
PromotionCategory	Category grouping (e.g., Reseller, Customer)
StartDate	Promotion effective start date
EndDate	Promotion effective end date
MinQty	Minimum order quantity to qualify
MaxQty	Maximum order quantity for the promotion

4. Analytical Capabilities

The schema shall support the following analytical queries:

- Total sales, discounts, net sales, and margin by date
- Sales by product, category, and subcategory
- Sales by customer, customer type, and geography
- Sales by territory, country, and region
- Sales by promotion, promotion type, and promotion category
- Comparison of order date, ship date, and due date performance
- Identification of products covered by promotions
- Analysis of margin contribution by product and promotion

5. Relationships

All dimension-to-fact relationships are one-to-many. The model enforces the following:

Dimension	Fact Table	Role / Notes
DIM_DATE	FACT_PRODUCT_SALES	Order date, ship date, due date (role-playing)
DIM_CUSTOMER	FACT_PRODUCT_SALES	Customer of the sales transaction
DIM_PRODUCT	FACT_PRODUCT_SALES	Product sold in the transaction
DIM_TERRITORY	FACT_PRODUCT_SALES	Sales territory of the transaction
DIM_PROMOTION	FACT_PRODUCT_SALES	Promotion applied to the transaction
DIM_PRODUCT	FACT_PROMOTION_COVERAGE	Product eligible for a promotion
DIM_PROMOTION	FACT_PROMOTION_COVERAGE	Promotion covering a product

6. Summary of Grain

Fact Table	Grain Definition
FACT_PRODUCT_SALES	One row per sales order line item — a product sold to a customer in a territory with a promotion, tied to order, ship, and due dates.
FACT_PROMOTION_COVERAGE	One row per product–promotion eligibility combination.

7. Summary of Measures

Measure	Additivity	Fact Table
OrderQuantity	Additive	FACT_PRODUCT_SALES
GrossSalesAmount	Additive	FACT_PRODUCT_SALES
DiscountAmount	Additive	FACT_PRODUCT_SALES
NetSalesAmount	Additive	FACT_PRODUCT_SALES
StandardCostAmount	Additive	FACT_PRODUCT_SALES
MarginAmount	Additive	FACT_PRODUCT_SALES
UnitPrice	Semi-descriptive	FACT_PRODUCT_SALES
(count of rows)	Factless / Derived	FACT_PROMOTION_COVERAGE

Section 5: ETL (Flow)

1. SQL Task: Staging Area Truncation

The staging tables for Sales, Product, Customer, Territory, and Promotion data are truncated prior to loading to remove any existing records, ensuring that each ETL cycle begins with a clean state. This guarantees accurate population of the dimensional model (DIM_DATE, DIM_PRODUCT, DIM_CUSTOMER, DIM_TERRITORY, DIM_PROMOTION) and FACT_PROMOTION_COVERAGE preventing duplication and preserving data integrity across the star schema.

2. Loading Staging Tables

Objective

This report explains: why staging tables are used in the ETL pipeline, what occurs in each data flow shown, and the source and destination databases involved.

Purpose of the Staging Area

The staging area is an intermediate layer where raw data is extracted from source systems and stored temporarily before transformation and loading into dimensional models.

Rationale

Isolation of source systems: Data is extracted once and processed independently, preventing repeated load on operational databases. (Source: Kimball, R. & Caserta, J., The Data Warehouse ETL Toolkit, Wiley, Chapter 5.)

Data validation and cleansing checkpoint: Staging allows detection of missing values, duplicates, and inconsistencies before loading into dimensions and facts. (Source: Inmon, W.H., Building the Data Warehouse, Wiley, Chapter 7.)

Support for complex transformations: Transformations such as joins, calculations, and surrogate key assignments are performed efficiently within the warehouse environment. (Source: Kimball Group, "ETL Architecture," <https://www.kimballgroup.com>)

Auditability and recovery: Raw data snapshots enable reprocessing in case of failures or logic changes. (Source: Golfarelli, M. & Rizzi, S., Data Warehouse Design: Modern Principles and Methodologies, McGraw-Hill, Chapter 6.)

High-Level Process Flow

The ETL workflow shown performs: extraction from source systems, loading into staging tables (stg_*), and preparation for downstream loading into dimension and fact tables. Each DFT (Data Flow Task) represents a pipeline that copies data from a source database into a staging database.

Data Flow Breakdown

DFT_Load_stg_SalesTerritory

Source: OLTP / source database (e.g., AdventureWorks transactional DB)

Destination: Staging database (stg_SalesTerritory table)

Territory data (region, country codes, groups) is extracted and copied without transformation into staging.

DFT_Load_stg_SpecialOffer

Source: Source database (SpecialOffer table)

Destination: Staging database (stg_SpecialOffer)

Promotion definitions (discounts, types, date ranges) are extracted and stored as raw promotion data for later use in dimension loading.

DFT_Load_stg_Product

Source: Source database (Product-related tables)

Destination: Staging database (stg_Product)

Product attributes (name, category, cost, price) are extracted with no business logic applied yet.

DFT_Load_stg_Customer

Source: Source database (Customer, Person, Store tables)

Destination: Staging database (stg_Customer)

Customer master data is extracted, including both individuals and store accounts, and stored in raw form for later SCD processing.

DFT_Load_stg_PromotionCoverage

Source: Source database (relationship between products and promotions)

Destination: Staging database (stg_PromotionCoverage)

Extracts mapping of which products are eligible for which promotions. No filtering or aggregation applied.

DFT_Load_stg_Sales

Source: Source database (SalesOrderHeader and SalesOrderDetail)

Destination: Staging database (stg_Sales)

Transactional sales data is extracted at order-line level, including quantities, prices, and discounts. This is the raw input for the fact table.

3. Loading DIM_PRODUCT and DIM_CUSTOMER (SCD Type 2)

The loading of Product and Customer dimensions implements a **Slowly Changing Dimension Type 2 (SCD Type 2)** pattern, which is designed to preserve historical changes by creating new versions of records instead

of overwriting existing ones. This process is more complex than standard dimension loading because it must detect changes, manage record validity periods, and maintain both current and historical versions simultaneously.

The process begins with data extraction from the staging tables using the **OLE DB Source**. At this stage, the data represents the most recent snapshot of source system data that has already been copied into the staging layer. No history exists in staging; it only contains the latest state of each entity.

Immediately after extraction, a **Derived Column** transformation is applied (e.g., `DC_Product_Clean` or `DC_Customer_Clean`). This step standardizes the incoming data before comparison. Typical operations include trimming strings, replacing `NULL` values with defaults, normalizing text formats, and preparing fields that will be used in change detection. This step is critical because inconsistent formatting would lead to false change detection.

The next step is the **Lookup** transformation (e.g., `LKP_Product_Current`, `LKP_Customer_Current`). This lookup queries the target dimension table but only against the current active records (`IsCurrent = 1`). The join is performed using the business key:

- `ProductID` for the product dimension
- `CustomerID` for the customer dimension

The lookup produces two logical outputs:

1. **No Match Output (New Records)** If no existing record is found, the incoming row represents a completely new business entity. These rows are routed directly to an insert destination (`DEST_InsertNewProduct` or `DEST_InsertNewCustomer`). A new surrogate key is generated automatically, and the record is inserted with:
 - `IsCurrent = 1`
 - `EffectiveStartDate = current load date`
 - `EffectiveEndDate = 9999-12-31`
2. **Match Output (Existing Records)** If a match is found, the process proceeds to determine whether the record has changed.

The matched records are passed into a **Conditional Split** transformation (e.g., `CS_Product_Changes`, `CS_Customer_Changes`). This step compares each tracked attribute between the incoming record and the existing dimension record. Only attributes defined as historically tracked are evaluated (e.g., product attributes like color, size, category; customer attributes like location or type).

The Conditional Split divides the flow into two paths:

- **No Change Path** If all tracked attributes are identical, the record is ignored. No update or insert occurs. This prevents unnecessary duplication and preserves performance.
- **Changed Path** If any tracked attribute differs, the record is identified as a change and must follow the SCD Type 2 process.

For changed records, the process executes in two coordinated steps:

Step 1: Expire Existing Record

The current version of the record in the dimension table is updated using a command transformation (`CMD_Expire_Product` or `CMD_Expire_Customer`). This operation:

- sets `IsCurrent = 0`
- updates `EffectiveEndDate` to the current load timestamp

This marks the record as historical and closes its validity period.

Step 2: Create New Version

The incoming record is then passed through another **Derived Column** transformation (e.g., `DC_Product_NewVersion` , `DC_Customer_NewVersion`). This step prepares the new version of the record by assigning SCD metadata:

- `EffectiveStartDate = current load date`
- `EffectiveEndDate = 9999-12-31`
- `IsCurrent = 1`

The new version is then inserted into the dimension table via a destination component (`DEST_InsertChangedProduct` or `DEST_InsertChangedCustomer`). A new surrogate key is generated, making this version distinct from the previous one.

The final result is that the dimension table contains multiple rows for the same business key, each representing a different historical version of the entity. At any point in time:

- only one record per business key has `IsCurrent = 1`
- all previous versions are retained with `IsCurrent = 0` and closed date ranges

This design enables time-variant analysis. Fact tables can join to the correct version of a dimension record based on the transaction date, ensuring that historical reports reflect the attributes that were valid at the

time of the event.

From a data flow perspective, the process ensures:

- new entities are inserted without duplication
- unchanged entities are ignored efficiently
- changed entities generate both an update (expire) and an insert (new version)
- full history is preserved without overwriting

Overall, the SCD Type 2 implementation for Product and Customer dimensions introduces controlled complexity to guarantee historical accuracy, auditability, and correct temporal reporting across the data warehouse.

4. Subsequent Control Flow Steps

4.1 SQL_Insert_Unknown_Members

Inserts default “Unknown” rows into all major dimensions so fact records with failed lookups can still load using surrogate key -1 .

See **Section 6: Insertion of Unknown Members in Dimension Tables** for the full table-by-table logic and defaults.

4.2 SQL_Load_DIM_DATE

Populates the date dimension by generating calendar and fiscal attributes, creating reusable date records for all fact tables.

4.3 SQL_Load_FACT_PROMOTION_COVERAGE (Factless Fact Table)

The loading of the factless fact table (`FACT_PROMOTION_COVERAGE`) captures the relationship between products and promotions without storing measurable metrics. Unlike standard fact tables, this table does not contain quantitative values such as sales amount or quantity. Instead, it records the existence of a valid relationship between two dimensions: Product and Promotion.

The process begins with extracting data from the staging table (`stg_PromotionCoverage`) using the **OLE DB Source**. This staging table contains raw mappings between products and promotions as defined in the source system. Each row represents that a specific product is eligible for a specific promotion.

A **Derived Column** transformation may be applied to standardize or prepare the incoming data. This can include handling null values, ensuring correct data types, or preparing keys for lookup operations. Although

minimal transformation is required compared to dimensions or transactional facts, this step ensures consistency before key resolution.

The next critical step is surrogate key resolution using **Lookup** transformations. Since the staging data contains business keys (e.g., `ProductID` , `SpecialOfferID`), these must be translated into surrogate keys used in the data warehouse:

- A lookup is performed against `DIM_PRODUCT` to retrieve `ProductKey`
- A lookup is performed against `DIM_PROMOTION` to retrieve `PromotionKey`

These lookups ensure that the factless fact table references the correct dimension records, including the appropriate SCD Type 2 version where applicable. If a match is not found, the process typically assigns the default unknown key (`-1`) to maintain referential integrity.

After successful key resolution, the data flows into the **OLE DB Destination**, where records are inserted into `FACT_PROMOTION_COVERAGE` . Each inserted row contains:

- `ProductKey`
- `PromotionKey`

No measures are stored, as the purpose of this table is purely relational.

The grain of this table is defined as:

one row per Product–Promotion relationship

This means that each combination of product and promotion appears only once. Duplicate prevention is typically enforced either through upstream deduplication logic or database constraints.

This table serves several analytical purposes:

- Identifies which products are eligible for which promotions
- Enables comparison between eligible promotions and actually applied promotions (from the sales fact table)
- Supports analysis of promotion coverage versus promotion usage
- Allows filtering of products based on promotion availability

From a data flow perspective, the process is straightforward compared to SCD-based dimensions:

- Extract relationship data from staging
- Resolve surrogate keys through lookups

- Insert clean, deduplicated relationships into the factless fact table

The result is a lightweight but critical structure that supports many-to-many relationships between dimensions, enabling more advanced analytical scenarios without introducing redundant measures.

This factless fact table complements the main sales fact table by providing context about promotion eligibility, allowing the warehouse to distinguish between what could have been promoted and what was actually sold under promotion.

4.4 SQL_Load_FACT_PRODUCT_SALES (Main Fact Table)

The loading of the main fact table (`FACT_PRODUCT_SALES`) represents the final stage of the ETL pipeline, where fully prepared transactional data is transformed into an analytical structure by resolving all dimension relationships and computing business measures.

This process differs from dimension loading because it does not create or modify descriptive entities. Instead, it focuses on integrating transactional data with dimension surrogate keys and producing numeric measures for analysis.

The data flow begins with the **OLE DB Source**, which extracts sales data from the staging table (`stg_Sales`). This staging table contains detailed transactional data at the order line level, including order identifiers, product identifiers, customer identifiers, dates, quantities, and pricing information. At this point, all keys are still business keys, not surrogate keys.

The first set of transformations focuses on resolving date relationships. Three separate **Lookup** transformations are used:

- `LKP_OrderDate`
- `LKP_DueDate`
- `LKP_ShipDate`

Each lookup maps a date value from the staging data to a corresponding `DateKey` in `DIM_DATE` . Since the date dimension is pre-populated, this step ensures that all temporal references in the fact table use consistent surrogate keys. If a date is missing or invalid, the lookup typically assigns the default unknown key (`-1`).

After resolving dates, the flow proceeds to resolve product and customer relationships:

- `LKP_Product` maps `ProductID` → `ProductKey`
- `LKP_Customer` maps `CustomerID` → `CustomerKey`

These lookups are especially important because both Product and Customer dimensions use SCD Type 2. This means the lookup must return the correct version of the dimension record based on the transaction context. Typically, the lookup is configured to match on business key and filter for the current record (`IsCurrent = 1`), unless a historical lookup strategy is implemented.

Next, the process resolves the geographic context:

- `LKP_Territory` maps territory-related identifiers to `TerritoryKey` in `DIM_TERRITORY`

This ensures that each sales record is associated with the correct region for analytical slicing.

Following this, the promotion relationship is resolved:

- `LKP_Promotion` maps `SpecialOfferID` → `PromotionKey`

This step links each sales transaction to the promotion that was applied, if any. If no valid promotion exists, the unknown key (`-1`) is used.

Once all foreign keys are resolved, the process performs a lookup against the fact table itself:

- `LKP_FactExisting`

This lookup checks whether the same sales record already exists in the fact table, typically using a combination of business identifiers such as `SalesOrderID` and `SalesOrderDetailID`. This step prevents duplicate inserts and ensures idempotent loading behavior.

Only records that do not already exist proceed further in the pipeline.

The next step is the **Derived Column** transformation (`DC_FinalMeasures`), where all numerical measures required for analysis are calculated. These measures are derived from the raw transactional fields and include:

- `GrossSalesAmount = OrderQuantity × UnitPrice`
- `DiscountAmount` = calculated based on discount percentage or discount value
- `NetSalesAmount = GrossSalesAmount - DiscountAmount`
- `StandardCostAmount = OrderQuantity × StandardCost`
- `MarginAmount = NetSalesAmount - StandardCostAmount`

This step ensures that all business metrics are precomputed and stored in the fact table, improving query performance and consistency.

Finally, the processed data is loaded into the **OLE DB Destination**, which inserts records into the `FACT_PRODUCT_SALES` table in the data warehouse.

Each inserted row contains:

- Foreign keys:
 - `OrderDateKey` , `ShipDateKey` , `DueDateKey`
 - `ProductKey` , `CustomerKey` , `TerritoryKey` , `PromotionKey`
- Degenerate dimensions:
 - `SalesOrderID` , `SalesOrderDetailID`
- Measures:
 - `OrderQuantity` , `UnitPrice`
 - `GrossSalesAmount` , `DiscountAmount` , `NetSalesAmount`
 - `StandardCostAmount` , `MarginAmount`

The grain of the fact table is strictly defined as:

one row per sales order line (`SalesOrderID` + `SalesOrderDetailID`)

This ensures that all measures are stored at the correct level of detail and can be aggregated accurately.

From a process perspective, this data flow guarantees:

- all business keys are converted into surrogate keys
- referential integrity is maintained across all dimensions
- duplicate records are avoided
- all required analytical measures are computed before loading
- the fact table is fully aligned with the dimensional model

Overall, this step integrates all previously prepared components—dimensions, staging data, and business rules—into a single, query-optimized fact table that supports comprehensive analytical reporting.

5. Source and Destination Databases

Layer	Database Type	Description
Source	OLTP / transactional database	Contains operational data (orders, customers, products)
Staging	Data warehouse staging database	Stores raw extracted data (stg_* tables)
Target (not shown fully here)	Data warehouse (dimensional model)	Contains DIM_* and FACT_* tables

Section 6: Insertion of Unknown Members in Dimension Tables

1. Objective

This process ensures that all dimension tables contain a default **“Unknown” member** with surrogate key -1 . This record is used when a fact record cannot be matched to a valid dimension entry.

2. Purpose of Unknown Members

Unknown members are required to maintain referential integrity between fact and dimension tables.

Key purposes:

- Prevent ETL failures when lookups fail
- Allow fact data to load even with missing dimension references
- Provide consistent reporting by grouping missing data under “Unknown”
- Ensure no data loss during ETL processing

3. General Logic Pattern

Each block follows the same structure:

1. Check if an unknown record already exists
2. If not, insert a default row

3. Use surrogate key -1
4. Populate attributes with neutral or placeholder values

4. Table-by-Table Breakdown

4.1 DIM_DATE

Condition:

- Check if `DateKey = -1` exists

Action:

- Insert a default date record

Key characteristics:

- `DateKey = -1`
- `FullDate = '1900-01-01'`
- `Text attributes = 'Unknown'`
- `Numeric attributes = default values`
- `Special fields:`
 - `QuarterName = 'U'`
 - `YearMonthKey = -1`

Purpose:

- Handles missing or invalid date references in fact data

4.2 DIM_TERRITORY

Condition:

- Check if `TerritoryKey = -1` or `TerritoryID = -1`

Action:

- Insert unknown territory record

Key characteristics:

- Keys set to -1
- TerritoryName = 'Unknown'
- CountryRegionCode = 'UNK'
- RegionGroup = 'Unknown'

Special handling:

- Uses IDENTITY_INSERT to allow manual key insertion

4.3 DIM_PROMOTION

Condition:

- Check if PromotionKey = -1 or SpecialOfferID = -1

Action:

- Insert unknown promotion record

Key characteristics:

- DiscountPct = 0
- Dates default to '1900-01-01' or NULL
- Quantity limits = 0 or NULL
- All descriptive fields = 'Unknown'

Purpose:

- Handles sales records without valid promotion data

4.4 DIM_PRODUCT

Condition:

- Check if ProductKey = -1 or ProductID = -1

Action:

- Insert unknown product record

Key characteristics:

- Descriptive attributes = 'Unknown'
- Costs and prices = 0
- Lifecycle dates = NULL
- SCD fields:
 - EffectiveStartDate = '1900-01-01'
 - EffectiveEndDate = '9999-12-31'
 - IsCurrent = 1

Purpose:

- Ensures fact rows with missing product references can still load

4.5 DIM_CUSTOMER**Condition:**

- Check if CustomerKey = -1 or CustomerID = -1

Action:

- Insert unknown customer record

Key characteristics:

- CustomerName = 'Unknown Customer'
- Identity and location fields = 'Unknown'
- AccountNumber = 'UNK'
- SCD fields:
 - EffectiveStartDate = '1900-01-01'
 - EffectiveEndDate = '9999-12-31'
 - IsCurrent = 1

Purpose:

- Handles missing or unresolved customer references

5. Role of IDENTITY_INSERT

Used to allow explicit insertion of surrogate key -1 into identity columns.

- Enabled before insert
- Disabled after insert
- Ensures system-generated keys are not disrupted

6. Execution Behavior

- Each block executes independently
- Inserts occur only if the record does not already exist
- Safe to rerun (idempotent process)

Section 7: OLAP Cube for Analytical Reporting

Cube Name

[AdventureWorksDW_View]

Purpose

This cube is designed to support analysis of product sales across time, product, customer, territory, and promotion dimensions. It exposes the core measures needed for revenue, discount, cost, and profitability analysis.

Measures

- [Measures].[Order Quantity]
- [Measures].[Gross Sales Amount]
- [Measures].[Discount Amount]
- [Measures].[Net Sales Amount]

- [Measures].[Margin Amount]

Main Dimensions

- **Order Date**

- Year
- Quarter
- Month
- Weekday
- Fiscal Year

- **Product**

- Product Name
- Subcategory Name
- Category Name

- **Customer**

- Customer Name
- Customer Type

- **Territory**

- Territory Name
- Country Region Code
- Region Group

- **Promotion**

- Promotion Type
- Promotion Category

MDX Queries for Business Goals

-- 1) Total sales measures by order year

```
SELECT
{
    [Measures].[Gross Sales Amount],
    [Measures].[Discount Amount],
    [Measures].[Net Sales Amount],
    [Measures].[Margin Amount]
} ON COLUMNS,
NON EMPTY
[Order Date].[Year Num].Members ON ROWS
FROM [AdventureWorksDW_View];
```

-- 2) Net sales by order quarter

```
SELECT
{ [Measures].[Net Sales Amount] } ON COLUMNS,
NON EMPTY
[Order Date].[Year Quarter Name].Members ON ROWS
FROM [AdventureWorksDW_View];
```

-- 3) Net sales by order month

```
SELECT
{ [Measures].[Net Sales Amount] } ON COLUMNS,
NON EMPTY
[Order Date].[Year Month Name].Members ON ROWS
FROM [AdventureWorksDW_View];
```

-- 4) Sales by product category and year

```
SELECT
[Order Date].[Year Num].Members ON COLUMNS,
NON EMPTY
[DIM PRODUCT].[Category Name].Members ON ROWS
FROM [AdventureWorksDW_View]
WHERE ([Measures].[Net Sales Amount]);
```

-- 5) Top 10 products by margin

```
SELECT
{ [Measures].[Margin Amount] } ON COLUMNS,
TOPCOUNT(
```

```

        NONEMPTY(
            [DIM PRODUCT].[Product Name].Members,
            [Measures].[Margin Amount]
        ),
        10,
        [Measures].[Margin Amount]
    ) ON ROWS
FROM [AdventureWorksDW_View];

-- 6) Top 10 products by net sales
SELECT
{ [Measures].[Net Sales Amount] } ON COLUMNS,
TOPCOUNT(
    NONEMPTY(
        [DIM PRODUCT].[Product Name].Members,
        [Measures].[Net Sales Amount]
    ),
    10,
    [Measures].[Net Sales Amount]
) ON ROWS
FROM [AdventureWorksDW_View];

-- 7) Sales by subcategory
SELECT
{
    [Measures].[Order Quantity],
    [Measures].[Net Sales Amount],
    [Measures].[Margin Amount]
} ON COLUMNS,
NON EMPTY
[DIM PRODUCT].[Subcategory Name].Members ON ROWS
FROM [AdventureWorksDW_View];

-- 8) Sales by customer type
SELECT
{
    [Measures].[Net Sales Amount],
    [Measures].[Margin Amount]
} ON COLUMNS,
NON EMPTY
[DIM CUSTOMER].[Customer Type].Members ON ROWS

```

```

FROM [AdventureWorksDW_View];

-- 9) Top customers by revenue
SELECT
{ [Measures].[Net Sales Amount] } ON COLUMNS,
TOPCOUNT(
    NONEMPTY(
        [DIM CUSTOMER].[Customer Name].Members,
        [Measures].[Net Sales Amount]
    ),
    10,
    [Measures].[Net Sales Amount]
) ON ROWS
FROM [AdventureWorksDW_View];

-- 10) Sales by country
SELECT
{
    [Measures].[Net Sales Amount],
    [Measures].[Margin Amount]
} ON COLUMNS,
NON EMPTY
[DIM TERRITORY].[Country Region Code].MEMBERS ON ROWS
FROM [AdventureWorksDW_View];

-- 11) Sales by region group
SELECT
{
    [Measures].[Net Sales Amount],
    [Measures].[Margin Amount]
} ON COLUMNS,
NON EMPTY
[DIM TERRITORY].[Region Group].Members ON ROWS
FROM [AdventureWorksDW_View];

-- 12) Promotion performance by promotion type
SELECT
{
    [Measures].[Gross Sales Amount],
    [Measures].[Discount Amount],
    [Measures].[Net Sales Amount],

```

```

        [Measures].[Margin Amount]
    } ON COLUMNS,
    NON EMPTY
    [DIM PROMOTION].[Promotion Type].Members ON ROWS
    FROM [AdventureWorksDW_View];

-- 13) Promotion performance by promotion category
SELECT
{
    [Measures].[Gross Sales Amount],
    [Measures].[Discount Amount],
    [Measures].[Net Sales Amount],
    [Measures].[Margin Amount]
} ON COLUMNS,
    NON EMPTY
    [DIM PROMOTION].[Promotion Category].Members ON ROWS
    FROM [AdventureWorksDW_View];

-- 14) Promotion performance over time
SELECT
[Order Date].[Year Num].Members ON COLUMNS,
    NON EMPTY
    [DIM PROMOTION].[Promotion Type].Members ON ROWS
    FROM [AdventureWorksDW_View]
    WHERE ([Measures].[Net Sales Amount]);

-- 15) Order quantity by promotion type
SELECT
{
    [Measures].[Order Quantity],
    [Measures].[Discount Amount]
} ON COLUMNS,
    NON EMPTY
    [DIM PROMOTION].[Promotion Type].Members ON ROWS
    FROM [AdventureWorksDW_View];

-- 16) Net sales by weekday
SELECT
{ [Measures].[Net Sales Amount] } ON COLUMNS,
    NON EMPTY
    [Order Date].[Weekday Name].Members ON ROWS

```

```

FROM [AdventureWorksDW_View];

-- 17) Fiscal year analysis
SELECT
{
    [Measures].[Net Sales Amount],
    [Measures].[Margin Amount]
} ON COLUMNS,
NON EMPTY
[Order Date].[Fiscal Year].Members ON ROWS
FROM [AdventureWorksDW_View];

-- 18) Top 5 territories by margin
SELECT
{ [Measures].[Margin Amount] } ON COLUMNS,
TOPCOUNT(
    NONEMPTY(
        [DIM TERRITORY].[Territory Name].Members,
        [Measures].[Margin Amount]
    ),
    5,
    [Measures].[Margin Amount]
) ON ROWS
FROM [AdventureWorksDW_View];

-- 19) Top 5 promotion types by net sales
SELECT
{ [Measures].[Net Sales Amount] } ON COLUMNS,
TOPCOUNT(
    NONEMPTY(
        [DIM PROMOTION].[Promotion Type].Members,
        [Measures].[Net Sales Amount]
    ),
    5,
    [Measures].[Net Sales Amount]
) ON ROWS
FROM [AdventureWorksDW_View];

-- 20) Year comparison matrix
SELECT
[Order Date].[Year Num].Members ON COLUMNS,

```

```
{  
    [Measures].[Gross Sales Amount],  
    [Measures].[Net Sales Amount],  
    [Measures].[Margin Amount]  
} ON ROWS  
FROM [AdventureWorksDW_View];
```

MDX Query's answered Business Question

The following table maps each MDX query in this section to the corresponding business question defined in **Section 2**, and summarizes what the query returns.

MDX Query	Business Question Addressed	What the Query Does
1	Q1 — Annual Sales Performance Overview	Returns yearly totals for Gross Sales, Discount, Net Sales, and Margin. Years are on rows, measures on columns, using <code>NON EMPTY</code> to suppress empty years.
2	Q2 — Quarterly Net Sales Trends	Returns Net Sales by quarter (<code>[Order Date].[Year Quarter Name]</code>). Quarters are listed on rows; the single measure is on columns.
3	Q3 — Monthly Net Sales Patterns	Returns Net Sales by month (<code>[Order Date].[Year Month Name]</code>). Months are listed on rows; the single measure is on columns.
4	Q4 — Product Category Performance by Year	Produces a matrix of Year (columns) × Product Category (rows). The <code>WHERE ([Measures].[Net Sales Amount])</code> slice ensures the cell values represent Net Sales for each category/year combination.
5	Q5 — Top 10 Products by Profit Margin	Returns the Top 10 products ranked by Margin. <code>TOPCOUNT</code> is applied over <code>NONEMPTY([Product Name],[Margin Amount])</code> so only products with margin are considered before ranking.
6	Q6 — Top 10 Products by Net Sales Revenue	Returns the Top 10 products ranked by Net Sales. <code>TOPCOUNT</code>

MDX Query	Business Question Addressed	What the Query Does
		is applied over NONEMPTY([Product Name], [Net Sales Amount]) to avoid ranking empty members.
7	Q7 — Sales and Margin by Product Subcategory	Returns Order Quantity, Net Sales, and Margin by Product Subcategory. Subcategories are on rows, measures on columns, enabling subcategory comparison across volume and profitability.
8	Q8 — Revenue and Margin by Customer Type	Returns Net Sales and Margin grouped by Customer Type. Customer types are on rows; measures are on columns to compare profitability across segments.
9	Q9 — Top 10 Customers by Net Sales	Returns the Top 10 customers ranked by Net Sales. Uses TOPCOUNT over NONEMPTY([Customer Name], [Net Sales Amount]) to rank only customers with revenue.
10	Q10 — Sales and Margin by Country	Returns Net Sales and Margin by Country Region Code. Country members are on rows with the two measures on columns for cross-country profitability comparison.

MDX Query	Business Question Addressed	What the Query Does
11	Q11 — Sales and Margin by Region Group	Returns Net Sales and Margin by Region Group (broader territory grouping). Region groups are on rows; measures are on columns.
12	Q12 — Promotion Performance by Promotion Type	Returns Gross Sales, Discount, Net Sales, and Margin grouped by Promotion Type. Useful for comparing promotion types across both revenue and profitability.
13	Q13 — Promotion Performance by Promotion Category	Returns Gross Sales, Discount, Net Sales, and Margin grouped by Promotion Category to evaluate broader promotion groupings.
14	Q14 — Promotion Type Performance Over Time	Produces a Year (columns) × Promotion Type (rows) matrix for Net Sales. The <code>WHERE ([Measures].[Net Sales Amount])</code> slice makes the cell values Net Sales across the year/type combinations.
15	Q15 — Order Quantity and Discount by Promotion Type	Returns Order Quantity and Discount Amount by Promotion Type, enabling a volume-vs-discount cost comparison across promotion types.
16	Q16 — Net Sales by Day of Week	Returns Net Sales grouped by Weekday Name. Weekdays are on rows, making intra-week sales patterns easy to observe.

MDX Query	Business Question Addressed	What the Query Does
17	Q17 — Net Sales and Margin by Fiscal Year	Returns Net Sales and Margin by Fiscal Year to align analysis with fiscal reporting periods rather than calendar years.
18	Q18 — Top 5 Territories by Profit Margin	Returns the Top 5 territories ranked by Margin. Uses <code>TOPCOUNT over NONEMPTY([Territory Name], [Margin Amount])</code> to rank only territories with margin.
19	Q19 — Top 5 Promotion Types by Net Sales	Returns the Top 5 promotion types ranked by Net Sales. Uses <code>TOPCOUNT over NONEMPTY([Promotion Type], [Net Sales Amount])</code> to rank only types with Net Sales.
20	Q20 — Year-over-Year Sales Comparison Matrix	Returns a Year comparison matrix with Years on columns and selected measures (Gross Sales, Net Sales, Margin) on rows. This is a transposed/compact view for side-by-side year comparison.

Section 8: Power BI Dashboard (Live SSAS Connection)

2.3.1 OLAP Cube Connection Setup

The Power BI report is connected to the OLAP cube through a live connection to SQL Server Analysis Services (SSAS). To establish the connection, Power BI Desktop was opened and the Get Data option was selected, followed by choosing Analysis Services as the data source. The SSAS server name was entered and the Connect live option was selected to avoid importing data locally. The AdventureWorksDW_View database was then selected as the target cube and the connection was confirmed. A live connection means that no

data is physically imported into Power BI — all queries are executed directly against the cube in real time, ensuring that all visuals and reports always reflect the current state of the data warehouse.

2.3.2 Report Pages

Report Page 1 — Sales Overview

- **Visuals:** Card KPIs displaying Net Sales Amount, Gross Sales Amount, and Margin Amount; a line chart showing sales trends over time; and a column chart breaking down sales by product category.
- **Fields Used:** Net Sales Amount, Gross Sales Amount, Margin Amount — sliced by Order Date (Year, Month) and Product Category.
- **Business Insight:** Provides a high-level overview of overall business performance, allowing stakeholders to monitor revenue trends, profitability levels, and product category contribution at a glance.

Report Page 2 — Product Analysis

- **Visuals:** A bar chart ranking the top 10 products by Net Sales Amount; a treemap showing revenue contribution by subcategory; and a scatter plot illustrating the relationship between Discount Amount and Margin Amount.
- **Fields Used:** Net Sales Amount, Margin Amount, Discount Amount — sliced by Product Name and Subcategory Name.
- **Business Insight:** Identifies the highest-revenue products and exposes the trade-off between discounting and profitability, directly supporting product mix optimization and pricing strategy decisions.

Report Page 3 — Customer & Geography Analysis

- **Visuals:** A bar chart comparing sales by customer type; a ranked table of top customers by revenue; and a map or bar chart displaying sales performance by country and regional group.
- **Fields Used:** Net Sales Amount, Margin Amount — sliced by Customer Type, Customer Name, Country Region, and Region Group.
- **Business Insight:** Reveals which customer segments and geographic markets drive the most value, enabling targeted marketing strategies and informed regional resource allocation decisions.

Report Page 4 — Promotion Analysis

- **Visuals:** A column chart comparing sales performance by promotion type; a stacked chart showing promotion performance trends over time; and a comparison chart contrasting Discount Amount, Net Sales Amount, and Margin Amount side by side.
- **Fields Used:** Net Sales Amount, Discount Amount, Margin Amount — sliced by Promotion Type, Promotion Category, and Order Date.

- **Business Insight:** Evaluates the effectiveness of different promotional strategies, identifying which promotion types generate incremental revenue and which erode profitability through excessive discounting.

2.3.3 Key Insights & Findings

- **Insight 1 — Sales Concentration in Top Products:** A small subset of products accounts for the majority of total revenue, consistent with a Pareto distribution. This suggests that inventory investment and marketing efforts should be focused on sustaining and expanding the performance of these high-contributing products.
- **Insight 2 — High Discounts Do Not Guarantee Higher Margins:** Scatter analysis reveals that increased discounting frequently reduces overall profitability rather than generating incremental value. Pricing strategies should be reviewed to avoid excessive discount levels that erode margin without proportionally increasing volume.
- **Insight 3 — Significant Regional Performance Variation:** Certain territories and regions consistently outperform others in both revenue and margin. Resource allocation, sales team deployment, and marketing budgets should be directed toward high-performing markets while underperforming regions are investigated for growth opportunities.
- **Insight 4 — Promotion Effectiveness Varies by Type:** Some promotion categories generate strong sales volume but deliver low margin returns. Promotion strategy should be evaluated on the basis of return on investment rather than revenue impact alone, prioritizing promotions that balance volume growth with margin preservation.
- **Insight 5 — Strong Seasonal Demand Patterns:** Sales data reveals consistent peaks during specific months and quarters throughout the year. Inventory planning, staffing levels, and promotional timing should be aligned with these recurring seasonal demand cycles to maximize operational efficiency.
- **Insight 6 — Operational Delays Visible Through Date Dimension Analysis:** Differences between order date, ship date, and due date perspectives reveal gaps in fulfillment timing. These patterns indicate opportunities to improve logistics processes, reduce delivery lag, and better align operational capacity with demand peaks.